

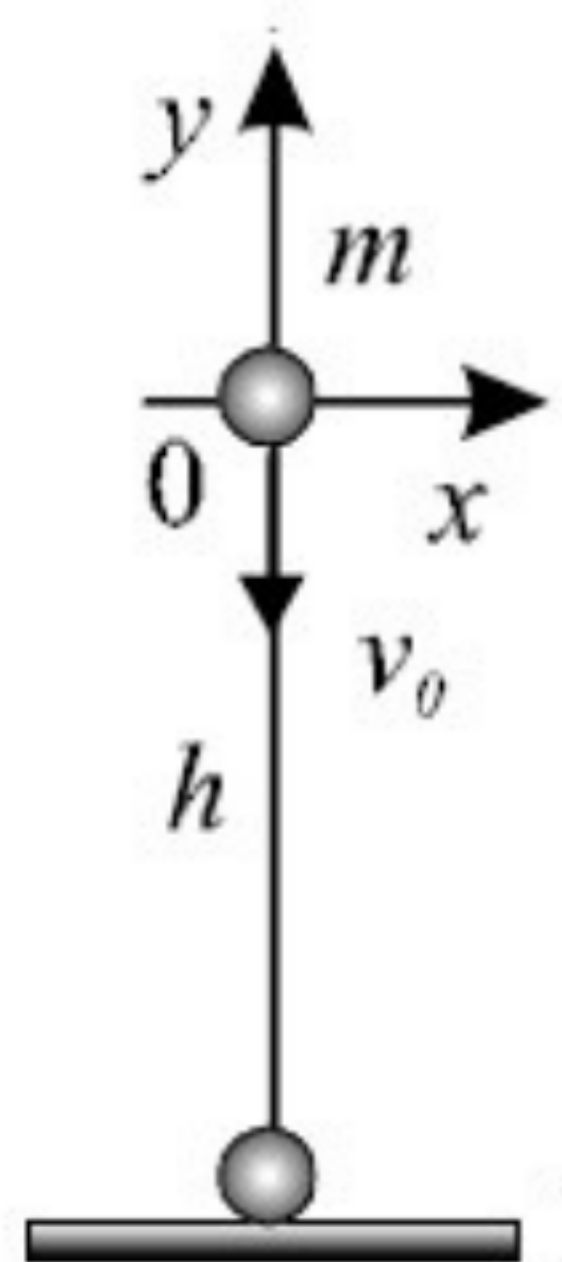
Ligjerata 3ç

Hedhja normale

$$v_0 \neq 0$$

$$\vec{v} = \vec{v}_0 + \vec{v}_g$$

$$\vec{v}_g = \vec{g}t$$



$$v = v_0 + gt \text{ sh. ecastit.}$$

$$v = \frac{ds}{dt} = \frac{dh}{dt}$$

$$\frac{dh}{dt} = v_0 + gt \text{ / } dt$$

$$dh = v_0 dt + gtdt \text{ / } \int$$

$$h = v_0 t + g \frac{t^2}{2} + c$$

$$t=0, h=0, c=0$$

$$h = v_0 t + g \frac{t^2}{2}$$

$$v = v_0 + gt \rightarrow t = \frac{v - v_0}{g}$$

$$h = v_0 \left(\frac{v - v_0}{g} \right) + \frac{g}{2} \left(\frac{v - v_0}{g} \right)^2$$

$$h = \frac{v_0 v - v_0^2}{g} + \frac{g}{2} \frac{v^2 - 2v v_0 + v_0^2}{g^2}$$

$$h = \frac{\cancel{2v_0 v} - \cancel{2v_0^2} + v^2 - \cancel{2v v_0} + \cancel{v_0^2}}{2g}$$

$$h = \frac{v^2 - v_0^2}{2g} \rightarrow 2hg = v^2 - v_0^2$$

$$v^2 = v_0^2 + 2gh \text{ / } \sqrt{\quad}$$

$$v = \sqrt{v_0^2 + 2gh}$$